Tutorial 5

MAT3007 Optimization, 2023 Summer

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Outline

1. Simplex Tableau
2. Degeneracy
3. Two-Phase Simplex Method
4. Big-M Method

Review: Simplex Tableau

The simplex tableau is a table with the following structure when the basis is B (the corresponding basis matrix and objective coefficients are A_B and $\boldsymbol{c}_B$):

$\boldsymbol{c}^T - \boldsymbol{c}_B^T A_B^{-1} A$	$-\boldsymbol{c}_B^T A_B^{-1} \boldsymbol{b}$
$A_B^{-1} A$	$A_B^{-1} \boldsymbol{b}$

- The lower part is simply a transformation of the constraints. In particular, the right hand side column is the current basic variables.
- The upper left part is the reduced costs, and the upper right corner is the negative of the objective value.

Therefore, The simplex tableau should look like (after reordering the columns)

$\mathbf{0}_m$	$\boldsymbol{c}_N^T - \boldsymbol{c}_B^T A_B^{-1} A_N$	$-\boldsymbol{c}_B^T \boldsymbol{x}_B$
$\mathbf{I}_m$	$A_B^{-1} A_N$	$\boldsymbol{x}_B$

Here $\mathbf{0}_m$ is a vector of m zeros and $\mathbf{I}_m$ is the m-dimensional identity matrix.

The simplex tableau will maintain a canonical form for each iteration until we reach optimality. Each iteration consists of

- Determine optimality: look at the top row (the reduced costs)
  - If all nonnegative, then the solution is already optimal
  - Otherwise, choose the smallest index j such that $\overline{c}_j < 0$ to be the incoming basis (pivot column)
- Perform the minimal ratio test (between the current basic variables and the incoming column — positive entries only).
- Choose the row that attains the smallest ratio to be the pivot row, and determine the outgoing basis (smallest index rule)
- Update the tableau
  - Divide each entry in the pivot row by the pivot element.
  - Add proper multiples of the new pivot row to each other rows, such that all other elements in the pivot column become zeros (including the top row).

Exercise 1

Consider the following tableau

B	β	0	0	0	δ	2	0
2	-1	1	0	0	η	0	1
4	-1	0	0	1	-1	1	2
3	α	0	1	0	1	1	γ

The entries of $\alpha, \beta, \eta, \delta$ and γ are unknown parameters. Find sufficient conditions on the range of them such that the following statements are true.

1. The current basic solution is optimal for the problem
2. The current basic solution is feasible. At the next iteration, x_5 is the only candidate to enter the basis. And if x_5 enters the basis, x_2 leaves the basis according to Bland's rule.
3. The current basic solution is feasible, but the problem is unbounded.
4. The last row indicates that the problem is infeasible.

Solution to Exercise 1

1. The current basic solution is optimal for the problem.

$$\beta \geq 0, \delta \geq 0, \gamma \geq 0$$

2. The current basic solution is feasible. At the next iteration, x_5 is the only candidate to enter the basis. And if x_5 enters the basis, x_2 leaves the basis according to Bland's rule.

$$\delta < 0, \beta \geq 0, 1/\eta \leq \gamma, \eta > 0$$

3. The current basic solution is feasible, but the problem is unbounded.

$$\beta < 0, \alpha \leq 0, \gamma \geq 0$$

4. The last row indicates that the problem is infeasible.

$$\alpha \geq 0, \gamma < 0$$

Exercise 2

Use the simplex method to solve the following problem

$$\begin{aligned} \max \quad & 3x_1 + 4x_2 \\ s.t. \quad & x_1 + x_2 \leq 4 \\ & 2x_1 + x_2 \leq 5 \\ & x \geq 0 \end{aligned} \tag{1}$$

Solution to Exercise 2

Table: Iteration 1

B	-3	-4	0	0	0
3	1	1	1	0	4
4	2	1	0	1	5

- By Bland's rule, bring index 1 in. Pivot column is column 1.
- By MRT, $\min\{\frac{4}{1}, \frac{5}{2}\} = 2.5$, The second row is pivot row and the outgoing index is 5.
- Use pivot entry $(2, 1)$ to do row combination and change all other entries on column 1 to be 0.

Solution to Exercise 2

Table: Iteration 2

B	0	$-\frac{5}{2}$	0	$\frac{3}{2}$	$\frac{15}{2}$
3	0	$\frac{1}{2}$	1	$-\frac{1}{2}$	$\frac{3}{2}$
1	1	$\frac{1}{2}$	0	$\frac{1}{2}$	$\frac{5}{2}$

- By Bland's rule, bring index 2 in. Pivot column is column 2.
- By MRT, $\min\{3, 5\} = 3$, The first row is pivot row and the outgoing index is 3.
- Use pivot entry $(2, 4)$ to do row combination and change all other entries in column 2 to be 0.

Solution to Exercise 2

Table: Iteration 3

B	0	0	5	-1	15
2	0	1	2	-1	3
1	1	0	-1	1	1

- By Bland's rule, bring index 4 in. Pivot column is column 4.
- By MRT, the only $-d_1 = 1 > 0$. The second row is pivot row and the outgoing index is 1.
- Use pivot entry $(1, 2)$ to do row combination and change all other entries on column 1 to be 0.

Solution to Exercise 2

Table: Iteration 3

B	1	0	4	0	16
2	1	0	1	0	4
4	1	0	-1	1	1

- All reduced costs are nonnegative, this is the optimal solution.
- Optimal value is 16, optimal solution $x = [0, 4, 0, 1]^T$

Outline

1 Simplex Tableau

2 Degeneracy

3 Two-Phase Simplex Method

4 Big-M Method

Review: Degenerate BFS and Pivoting Rules

Definition 1 (Degeneracy).

We call a basic feasible solution x (non)degenerate if (none) some of the basic values are 0.

Theorem 2 (Bland's Rule).

If we use smallest index rule for choosing both the entering basis and the exiting basis, then no cycle will occur in the simplex algorithm.

Smallest index rule: Choose the smallest index j with $\overline{c}_j < 0$

Exercise 3

Use the simplex method to solve the following problem

$$
\begin{aligned}
\max \quad & 10x_1 - 57x_2 - 9x_3 - 24x_4 \\
\text{s.t.} \quad & 1/2x_1 - 11/2x_2 - 5/2x_3 + 9x_4 \leq 0 \\
& 1/2x_1 - 3/2x_2 - 1/2x_3 + x_4 \leq 0 \\
& x_1 + x_2 + x_3 + x_4 \leq 1 \\
& x_1, x_2, x_3, x_4 \geq 0
\end{aligned}
$$

Solution to Exercise 3

Table: Iteration 1

B	-10	57	9	24	0	0	0	0
5	$\frac{1}{2}$	$-\frac{11}{2}$	$-\frac{5}{2}$	9	1	0	0	0
6	$\frac{1}{2}$	$-\frac{3}{2}$	$-\frac{1}{2}$	1	0	1	0	0
7	1	1	1	1	0	0	1	1

Solution to Exercise 3

Table: Iteration 2

B	0	27	-1	44	0	20	0	0
5	0	-4	-2	8	1	-1	0	0
1	1	-3	-1	2	0	2	0	0
7	0	4	2	-1	0	-2	1	1

Solution to Exercise 3

Table: Iteration 3

B	0	29	0	$\frac{87}{2}$	0	19	$\frac{1}{2}$	$\frac{1}{2}$
5	0	0	0	7	1	-3	1	1
1	1	-1	0	$\frac{3}{2}$	0	1	$\frac{1}{2}$	$\frac{1}{2}$
3	0	2	1	$-\frac{1}{2}$	0	-1	$\frac{1}{2}$	$\frac{1}{2}$

- All reduced costs are nonnegative, this is the optimal solution.
- Optimal value is $\frac{1}{2}$, optimal solution $x = [\frac{1}{2}, 0, \frac{1}{2}, 0, 5, 0, 0]^T$

Solution to Exercise 3

$$
\begin{array}{lll}
\text{maximize} & z = 10x_1 - 57x_2 - 9x_3 - 24x_4 & \\
\text{subject to} & 1/2x_1 - 11/2x_2 - 5/2x_3 + 9x_4 \le 0 & \\
& 1/2x_1 - 3/2x_2 - 1/2x_3 + x_4 \le 0 & \\
& x_1 + x_2 + x_3 + x_4 \le 1 & \\
& x_1, x_2, x_3, x_4 \ge 0 &
\end{array}
$$

Initial tableau:

z	x_1	x_2	x_3	x_4	s_1	s_2	s_3	rhs		
1	−10	57	9	24	0	0	0	0	=	z
0	1/2	−11/2	−5/2	9	1	0	0	0	=	s_1
0	1/2	−3/2	−1/2	1	0	1	0	0	=	s_2
0	1	1	1	1	0	0	1	1	=	s_3

x_1 enters, s_1 leaves (in a tie for smallest ratio).

z	x_1	x_2	x_3	x_4	s_1	s_2	s_3	rhs		
1	0	−53	−41	204	20	0	0	0	=	z
0	1	−11	−5	18	2	0	0	0	=	x_1
0	0	4	2	−8	−1	1	0	0	=	s_2
0	0	12	6	−17	−2	0	1	1	=	s_3

x_2 enters (since $-53 < -41$), s_2 leaves

One method of preventing cycling is to use Bland's Rule instead of the Most Negative Coefficient Rule. Given a fixed ordering of the variables, e.g. $x_1, x_2, \ldots, x_n, s_1, \ldots, s_m$, Bland's Rule says:

(a) When choosing the entering variable, take the first one in the ordering that has a negative entry in the objective row.

(b) When choosing the leaving variable, if there is a tie for least ratio, take the candidate that is first in the ordering.

If we were using Bland's Rule, everything would have been the same up to that last pivot, where x_1 would have entered instead of s_2, and x_4 would have left.

z	x_1	x_2	x_3	x_4	s_1	s_2	s_3	rhs		
1	0	27	−1	44	0	20	0	0	=	z
0	0	−4	−2	8	1	−1	0	0	=	s_1
0	1	−3	−1	2	0	2	0	0	=	x_1
0	0	4	2	−1	0	−2	1	1	=	s_3

Then x_3 would have entered, and s_3 left. This would be, at last, a nondegenerate pivot, producing an optimal tableau:

z	x_1	x_2	x_3	x_4	s_1	s_2	s_3	rhs		
1	0	29	0	87/2	0	19	1/2	1/2	=	z
0	0	0	0	7	1	−3	1	1	=	s_1
0	1	−1	0	3/2	0	1	1/2	1/2	=	x_1
0	0	2	1	−1/2	0	−1	1/2	1/2	=	x_3

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Review: Two-Phase Method in Simplex Tableau

Phase I:

- The auxiliary problem is not in a canonical form: Although the constraint contains an identity matrix (which is the initial basis), the corresponding objective coefficients are not 0's
- We need to compute the top row (reduced costs)
  - For basic part, the reduced costs are 0
  - For nonbasic part, $\bar{c}_j = c_j - \boldsymbol{c}_B^T A_B^{-1} A_j = -\boldsymbol{e}^T A_j$, so the jth reduced cost is the negative of the sum of that column
- This also applies to the initial objective value, which equals the negative of the sum of the right hand side vector.

Phase II:

- Drop all the columns associated with the auxiliary variables
- Change the top row to the reduced costs corresponding to the original problem by using the reduced cost formula:

$$\bar{c}_j = c_j - \boldsymbol{c}_B^T A_B^{-1} A_j$$

  Also compute the current objective value
- Continue the simplex tableau to solve the original problem

If the optimal solution to the auxiliary problem is degenerate, then replace the basic indices for the auxiliary variables with (any) indices of original variables and recalculate the table.

Exercise 3: Two-phased Method

Use the two-phase simplex method to completely solve the linear optimization problem:

$$
\begin{aligned}
\min \quad & 2x_1 + 3x_2 + 3x_3 + x_4 - 2x_5 \\
s.t. \quad & x_1 + 3x_2 + \quad + 4x_4 + x_5 = 2 \\
& x_1 + 2x_2 + \quad - 3x_4 + x_5 = 2 \\
& x_1 + 4x_2 - 3x_3 + \quad + \quad = -1 \\
& x \geq 0
\end{aligned}
$$

Solution to Exercise 3: Phase 1

First, **make b positive** and construct the auxiliary problem:

$$
\begin{aligned}
\min \quad & y_1 + y_2 + y_3 \\
s.t. \quad & x_1 + 3x_2 + \quad + 4x_4 + x_5 + y_1 + \quad + \quad = 2 \\
& x_1 + 2x_2 + \quad - 3x_4 + x_5 + \quad + y_2 + \quad = 2 \\
& -x_1 - 4x_2 + 3x_3 + \quad + \quad + \quad + \quad + y_3 = 1 \\
& x, y \geq 0
\end{aligned}
$$

Solution to Exercise 3: Phase 1

Construct the initial tableau for the auxiliary problem.

Table: Iteration 1

B	-1	-1	-3	-1	-2	0	0	0	-5
6	1	3	0	4	1	1	0	0	2
7	1	2	0	-3	1	0	1	0	2
8	-1	-4	3	0	0	0	0	1	1

We bring x_1 into the basis and have x_6 exit the basis, the new tableau is:

Table: Iteration 2

B	0	2	-3	3	-1	1	0	0	-3
1	1	3	0	4	1	1	0	0	2
7	0	-1	0	-7	0	-1	1	0	0
8	0	-1	3	4	1	1	0	1	3

Solution to Exercise 3: Phase 1

We bring x_3 into the basis and have x_8 exit the basis, the new tableau is:

Table: Iteration 3

B	0	1	0	7	0	2	0	1	0
1	1	3	0	4	1	1	0	0	2
7	0	-1	0	-7	0	-1	1	0	0
3	0	$-\frac{1}{3}$	1	$\frac{4}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	0	$\frac{1}{3}$	1

We reach the optimality of auxiliary problem and the cost is zero while x_7 is still in the basis, indicates that we have a **feasible but not basic** solution to the original problem.
In order to obtain a basic solution, we need to drive x_7 out of the basis.
Note that x_7 is the 2nd basic variable, we examine the 2nd row of tableau and find j=2,4 that the 2nd of $A_B^{-1}A_j$ is nonzero. Choose j=2 as the index that enter the basis

Solution to Exercise 3: Phase 1

We bring x_2 into the basis and have x_7 exit the basis, **while keep the optimality**, the new tableau is:

Table: Iteration 4

B	0	0	0	0	0	1	1	1	0
1	1	0	0	-17	1	-2	3	0	2
2	0	1	0	7	0	1	-1	0	0
3	0	0	1	$\frac{11}{3}$	$\frac{1}{3}$	$\frac{2}{3}$	$-\frac{1}{3}$	$\frac{1}{3}$	1

Then we get a BFS of the original problem x=(2,0,1,0,0) and B=$\{1,2,3\}$
We drop all the columns for auxiliary variables. Then we recompute the reduced cost for the original problem.

Solution to Exercise 3: Phase 2

$x = (2, 0, 1, 0, 0), B = \{1, 2, 3\}$. We only need to calculate the reduced again:

$$\bar{c}^\top = c^\top - c_B^\top A_B^{-1} A = \begin{pmatrix} 2 & 3 & 3 & 1 & -2 \end{pmatrix} - \begin{pmatrix} 2 & 3 & 3 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & -17 & 1 \\ 0 & 1 & 0 & 7 & 0 \\ 0 & 0 & 1 & \frac{11}{3} & \frac{1}{3} \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 & 0 & 3 & -5 \end{pmatrix}$$

Table: Iteration 4

B	0	0	0	3	-5	-7
1	1	0	0	-17	1	2
2	0	1	0	7	0	0
3	0	0	1	$\frac{11}{3}$	$\frac{1}{3}$	1

Solution to Exercise 3: Phase 2

Table: Iteration 4

B	0	0	0	3	-5	-7
1	1	0	0	-17	1	2
2	0	1	0	7	0	0
3	0	0	1	$\frac{11}{3}$	$\frac{1}{3}$	1

Table: Iteration 5

B	5	0	0	-82	0	3
5	1	0	0	-17	1	2
2	0	1	0	7	0	0
3	$-\frac{1}{3}$	0	1	$\frac{11}{3}$	0	$\frac{1}{3}$

Solution to Exercise 3: Phase 2

Table: Iteration 5

B	5	0	0	-82	0	3
5	1	0	0	-17	1	2
2	0	1	0	7	0	0
3	$-\frac{1}{3}$	0	1	$\frac{11}{3}$	0	$\frac{1}{3}$

Table: Iteration 6

B	5	$\frac{82}{7}$	0	0	0	3
5	1	$\frac{17}{7}$	0	0	1	2
4	0	$\frac{1}{7}$	0	1	0	0
3	$-\frac{1}{3}$	$-\frac{4}{3}$	1	0	0	$\frac{1}{3}$

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Review: Big-M Method

Solve LP without a starting BFS. Consider the following auxiliary problem:

$$\begin{aligned} \text{minimize} \quad & \mathbf{c}^T\mathbf{x} + M\sum_{i=1}^{m} y_i \\ \text{subject to} \quad & A\mathbf{x} + \mathbf{y} = \mathbf{b} \\ & \mathbf{x}, \mathbf{y} \geq 0 \end{aligned}$$

This problem has an initial BFS $\mathbf{y} = \mathbf{b} \geq 0$ (again assuming $\mathbf{b} \geq 0$). Now we can use simplex to solve it. In the simplex procedure, pretend that M is a very large value (larger than any specified number).

Exercise 2: Big-M Method

Use the big-M simplex method to completely solve the linear optimization problem:

$$
\begin{aligned}
\min \quad & 4x_1 + x_2 + x_3 \\
s.t. \quad & 2x_1 + x_2 + 2x_3 = 4 \\
& 3x_1 + 3x_2 + x_3 = 3 \\
& x \geq 0
\end{aligned}
$$

Solution to Exercise 2

First, we use big-M method in conjunction with the following auxiliary problem:

$$\begin{aligned} \min \quad & 4x_1 + x_2 + x_3 + Mx_4 + Mx_5 \\ s.t. \quad & 2x_1 + x_2 + 2x_3 + x_4 + \quad = 4 \\ & 3x_1 + 3x_2 + x_3 + \quad + x_5 = 3 \\ & x \geq 0 \end{aligned}$$

Construct the initial tableau for the auxiliary problem with basis B={4,5} , c_B={M,M}, x=(0,0,0,4,3) and A_B is identity matrix.
Then $\bar{c} = c - A^T A_B^{-T} c_B = c - A^T c_B = (4 - 5M, 1 - 4M, 1 - 3M, 0, 0)^T$

Table: Iteration 1

B	4-5M	1-4M	1-3M	0	0	-7M
4	2	1	2	1	0	4
5	3	3	1	0	1	3

Solution to Exercise 2

The reduced cost of x_1 (i.e. 4-5M) is negative when M is large enough. We bring x_1 into the basis and have x_5 exit the basis, the new tableau is:

Table: Iteration 2

B	0	M-3	$-\frac{4}{3}M-\frac{1}{3}$	0	$\frac{5}{3}M-\frac{4}{3}$	-2M-4
4	0	-1	$\frac{4}{3}$	1	$-\frac{2}{3}$	2
1	1	1	$\frac{1}{3}$	0	$\frac{1}{3}$	1

The reduced cost of x_3 (i.e. $-\frac{4}{3}M-\frac{1}{3}$) is negative when M is large enough. We bring x_3 into the basis and have x_4 exit the basis, the new tableau is:

Table: Iteration 3

B	0	$-\frac{13}{4}$	0	M+$\frac{1}{4}$	$M-\frac{3}{2}$	$-\frac{7}{2}$
3	0	$-\frac{3}{4}$	1	$\frac{3}{4}$	$-\frac{1}{2}$	$\frac{3}{2}$
1	1	$\frac{5}{4}$	0	$-\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{2}$

Solution to Exercise 2

The reduced cost of x_2 (i.e. $-\frac{13}{4}$) is negative. We bring x_2 into the basis and have x_1 exit the basis, the new tableau is:

Table: Iteration 4

B	$\frac{13}{5}$	0	0	M-$\frac{2}{5}$	$M-\frac{1}{5}$	$-\frac{11}{5}$
3	$\frac{3}{5}$	0	1	$\frac{3}{5}$	$\frac{1}{5}$	$\frac{9}{5}$
2	$\frac{4}{5}$	1	0	$-\frac{1}{5}$	$\frac{2}{5}$	$\frac{2}{5}$

$\bar{c} \geq 0$ when M is large enough and we have an optimal solution to the auxiliary problem. In addition, all of the artificial variables have been driven to zero, so we have an optimal solution to the original problem.

$$x^* = (0, \frac{2}{5}, \frac{9}{5})$$

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